

# DehatiAI × Telenor Khushhaal Kisaan

## Financial Model, Profit Margin Summary & Unit Economics

### Confidential B2B Strategic Pricing & Revenue Distribution Framework

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### Executive Summary (Realistic Ground-Truth Model)

This document presents the realistic **Financial Architecture**, **Profit Margin Analysis**, and **Unit Economics** for embedding the **DehatiAI** engine into **Telenor Pakistan's Khushhaal Kisaan (7272)** digital ecosystem.

By leveraging Direct Carrier Billing (DCB) airtime deduction at an impulse price point of **PKR 10 / week (PKR 40 / month)**, this model maximizes rural adoption while delivering high net margins.

| REALISTIC GROUND-TRUTH HIGHLIGHTS |                                     |  |
|-----------------------------------|-------------------------------------|--|
| Subscription Pricing              | PKR 10 / Week (PKR 40 / Month)      |  |
| Target Paid Base (Year 1)         | 200,000 Active Paid Subscribers     |  |
| Year 1 Gross Revenue (ARR)        | PKR 96,000,000 (\$342,850 USD)      |  |
| Telenor Net Profit (60%)          | PKR 46,368,000 (\$165,600 USD / Yr) |  |
| DehatiAI Share (40%)              | PKR 30,912,000 (\$110,400 USD / Yr) |  |
| Infrastructure COGS Ratio         | ~3.8% of Gross Revenue              |  |
| Telenor Net Profit Margin         | 100% of Telenor Share (0 OpEx)      |  |

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### 1. Unit Economics Breakdown (Per Subscriber / Month)

Below is the micro-level breakdown of revenue, direct costs, and net margin per user based on **PKR 10 / week**.

| Financial Item           | PKR per User / Month | USD equivalent (\$1 = PKR 280) | % of Gross Revenue |
|--------------------------|----------------------|--------------------------------|--------------------|
| Gross Subscription Price | PKR 40.00            | \$0.143                        | 100.0%             |

|                                           |           |          |                         |
|-------------------------------------------|-----------|----------|-------------------------|
| Government Taxes / FED (19.5%)            | PKR 7.80  | \$0.028  | 19.5%                   |
| Net Collectible Revenue                   | PKR 32.20 | \$0.115  | 80.5%                   |
| Telenor Share (60% of Net)                | PKR 19.32 | \$0.069  | 48.3% Gross / 60% Net   |
| DehatiAI Share (40% of Net)               | PKR 12.88 | \$0.046  | 32.2% Gross / 40% Net   |
| Direct Operating COGS (DehatiAI Side):    |           |          |                         |
| • Anthropic Claude 3.5 API (Prompt Cache) | PKR 0.90  | \$0.0032 | 2.25%                   |
| • Cloud Hosting (Railway/PostgreSQL/CDN)  | PKR 0.22  | \$0.0008 | 0.56%                   |
| • Speech Engine & Twilio/SMS Fallback     | PKR 0.40  | \$0.0014 | 1.00%                   |
| Total COGS per User                       | PKR 1.52  | \$0.0054 | 3.81%                   |
| DehatiAI Net Profit per User              | PKR 11.36 | \$0.0406 | 88.2% of DehatiAI Share |
| Telenor Net Profit per User               | PKR 19.32 | \$0.0690 | 100% of Telenor Share   |

Zero COGS for Telenor: Telenor bears 0% infrastructure, server, or AI API cost. All technical overhead is absorbed by DehatiAI out of its 40% share.

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## ■ ■ 2. Revenue Share Models Comparison

We offer three flexible contract structures tailored for Telenor’s commercial leadership:

### Option A: Standard 60/40 VAS Revenue Share (Recommended)

- **Telenor Split:** 60% of Net Revenue (PKR 19.32 / user / month).
- **DehatiAI Split:** 40% of Net Revenue (PKR 12.88 / user / month).
- **Benefits:** Zero upfront setup fee for Telenor. Aligns incentives to scale subscriber base aggressively.

## Option B: B2B Enterprise SaaS License + Performance Royalty

- **Fixed Enterprise Fee:** PKR 6,000,000 (\$21,400 USD) per year for unlimited engine licensing.
- **Per User Royalty:** PKR 5 / subscriber / month.
- **Benefits:** Predictable software expenditure for Telenor with higher margins at large scale.

## Option C: Tiered Scale Split (Volume Incentive Model)

|                                   |                 |                  |   |
|-----------------------------------|-----------------|------------------|---|
| ■ Active Monthly Subscribers      | ■ Telenor Share | ■ DehatiAI Share | ■ |
| ■ Tier 1: 0 - 200,000 Subscribers | ■ 55%           | ■ 45%            | ■ |
| ■ Tier 2: 200,000 - 500,000 Subs  | ■ 60%           | ■ 40%            | ■ |
| ■ Tier 3: 500,000+ Subscribers    | ■ 65%           | ■ 35%            | ■ |

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### ■ 3. 3-Year Financial Projections (Realistic Ground-Truth Model)

Based on a conservative 2% conversion of Telenor’s 10M+ rural user base in Year 1 scaling to 10% in Year 3:

|                                                               |      |             |      |
|---------------------------------------------------------------|------|-------------|------|
| gantt                                                         |      |             |      |
| title Realistic Subscriber Growth Projection (3-Year Horizon) |      |             |      |
| dateFormat YYYY-MM-DD                                         |      |             |      |
| section Subscribers                                           |      |             |      |
| Y1 (200K Subs)                                                | :a1, | 2026-01-01, | 365d |
| Y2 (500K Subs)                                                | :a2, | 2027-01-01, | 365d |
| Y3 (1.0M Subs)                                                | :a3, | 2028-01-01, | 365d |

## Detailed Projections Table

| Metric (Realistic Model)         | Year 1          | Year 2          | Year 3           |  |
|----------------------------------|-----------------|-----------------|------------------|--|
| Active Subscribers (EOP)         | 200,000         | 500,000         | 1,000,000        |  |
| Average Monthly Paid Users       | 200,000         | 500,000         | 1,000,000        |  |
| Gross Subscription Revenue (PKR) | PKR 96,000,000  | PKR 240,000,000 | PKR 480,000,000  |  |
| Net Revenue after FED (PKR)      | PKR 77,280,000  | PKR 193,200,000 | PKR 386,400,000  |  |
| Telenor Earnings (60% Net)       | PKR 46,368,000  | PKR 115,920,000 | PKR 231,840,000  |  |
| (USD Equivalent @ 280)           | (\$165,600 USD) | (\$414,000 USD) | (\$828,000 USD)  |  |
| DehatiAI Earnings (40% Net)      | PKR 30,912,000  | PKR 77,280,000  | PKR 154,560,000  |  |
| Less: DehatiAI Infrastructure    | (PKR 3,648,000) | (PKR 9,120,000) | (PKR 18,240,000) |  |
| DehatiAI Net Profit              | PKR 27,264,000  | PKR 68,160,000  | PKR 136,320,000  |  |
| (USD Equivalent @ 280)           | (\$97,370 USD)  | (\$243,420 USD) | (\$486,850 USD)  |  |

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■ 4. Cost Optimization Strategy (Why COGS is only 3.8%)

DehatiAI achieves industry-leading profit margins through four proprietary architectural breakthroughs:

1. 4-Pillar Offline PWA Storage:
- 110+ Local Urdu FAQs run inside client-side IndexedDB (`dehati\_offline\_v2`).
  - 92% of common farmer queries (e.g., wheat water timings, fertilizer ratios) are answered locally on the device at **PKR 0.00 API cost**.
2. 2-Tier Prompt Caching (`aiCache.js`):
- High-frequency AI responses are cached in memory and PostgreSQL with a 3-second timeout fallback.
  - Reduces Anthropic API token expenditure by **68%**.
3. Client-Side Vision Compression:
- Farmers' leaf photos are compressed on the phone using `browser-image-compression` prior to upload, reducing server bandwidth cost by **85%**.
4. PyTorch ResNet-50 Local Inference:
- Standard plant disease scans bypass expensive LLM calls and route directly to lightweight local PyTorch vision model (`ResNet50-Plant-model-80.pth`).

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## ■ 5. Summary & Contractual Next Steps

- **Risk Profile: Zero Capital Expenditure for Telenor Pakistan.** DehatiAI hosts, maintains, updates, and secures the entire software stack.
- **Legal Agreement Type:** 3-Year Commercial B2B Enterprise & VAS Agreement.
- **Proposed Execution Date:** September 15, 2026.